Experiment 2: Text Recognition

Teaching a model to sort custody client messages into six intents from labelled examples, and Logistic Regression.

WHY THIS EXPERIMENT

In Experiment 1 rules failed on messages without the right keyword; here the model learns from labelled examples, so it can handle wording no one wrote a rule for.

USE CASE FOR NUVAMA

Every client email to Nuvama Asset Services is sorted into the right intent (and desk) automatically, so it reaches the right team without someone reading and forwarding it.

SAMPLE DATASET: "NUVAMA CUSTODY CLIENT QUERIES"

36 trainer-written custody client messages, 6 for each of 6 intents (14 easy, 12 medium, 10 hard, including Hinglish and deliberate traps), plus 12 unseen test messages: 10 with a correct intent and 2 that fit none.

The six intents in the data

Label (desk)	What it covers	Example query from the data	Why it matters to Nuvama
Statement request Client Reporting desk	Asks for a statement, report, contract note, NAV pack or other record of holdings and transactions.	*"Can you resend last week's contract notes? Our auditors are asking for them."*	Auditors, boards and fund accountants need holdings and transaction records on time.
Corporate action Corporate Actions desk	Dividends, bonus, rights, splits, buybacks, mergers, record dates and entitlements, including tax on them.	*"When will the HDFC Bank dividend be credited to our account?"*	Dividends, bonus, rights and buybacks must reach the client correctly by the record date.
Settlement status Settlement desk	Trades, pay-in and pay-out, delivery, short delivery, auctions, cut-off times and settlement codes.	*"Has yesterday's sale of 5,000 TCS shares settled?"*	Unsettled or short-delivered trades cost money and need action the same day.
Fees and billing Billing desk	Custody fees, invoices, GST on invoices, charges and fee schedules.	*"We were charged twice on the September invoice."*	Wrong charges or invoices hurt revenue and trust; they go to the billing team.
Onboarding and KYC Onboarding & KYC desk	FPI registration, account opening, KYC, signatories, UBO, FATCA/CRS and address changes.	*"Any update on our FPI registration?"*	The account cannot operate until registration, KYC and signatory details are complete.
Complaint or escalation Escalations desk	The client is unhappy, threatens to move, suspects fraud or asks for a senior person. Overrides the topic.	*"This is the third failed trade this month. This is unacceptable."*	An unhappy client, a threat to leave or suspected fraud needs a senior person, whatever the topic.

How the participant categorises the data

1 · READ
Each of the 36 queries is shown unlabelled, with its full text.

2 · CHOOSE
Pick one of the 6 labels from the dropdown; the guide says what each covers.

3 · COUNT
Counters show progress; at least 5 per label are needed to train.

4 · CHECK
“Check my labels” compares with the trainer’s answers and explains each difference.

5 · GROUP
“Group it” turns each label into a box of examples: the training data.

The traps are deliberate: “Your team missed our buyback tender. We want to speak to your head of operations.” mentions a corporate action but is a complaint, because the client asks for a senior person. Labelling teaches people that the intent, not the keyword, decides the label.

The two models

LOGISTIC REGRESSION
Counts the words and word pieces in each message and learns how strongly each one points to each label. Trains in seconds, works with few examples, easy to explain.

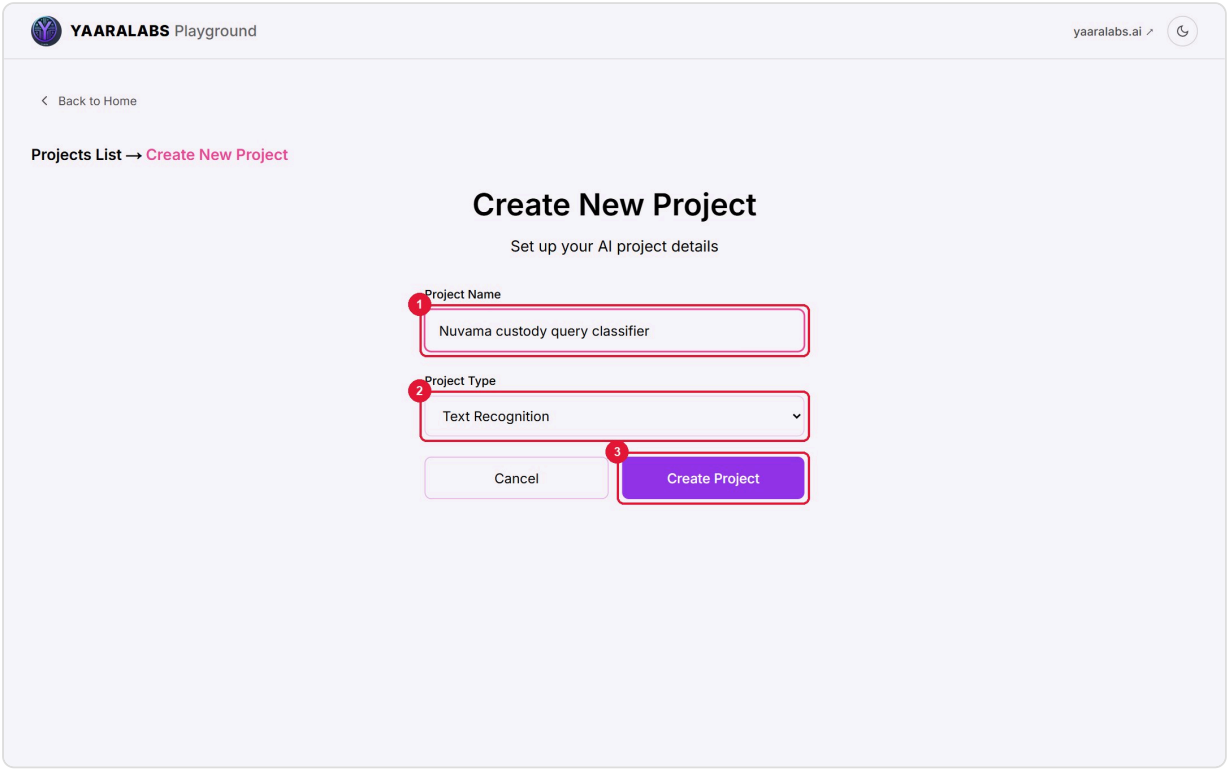
DISTILBERT (ALSO AVAILABLE)
A pretrained language model that reads the whole sentence and its context. It is the second option in the same model list; slower to train and usually needs more examples per label, so this walkthrough uses Logistic Regression.

Step by step

Red boxes mark what is typed or clicked; numbers show the order.

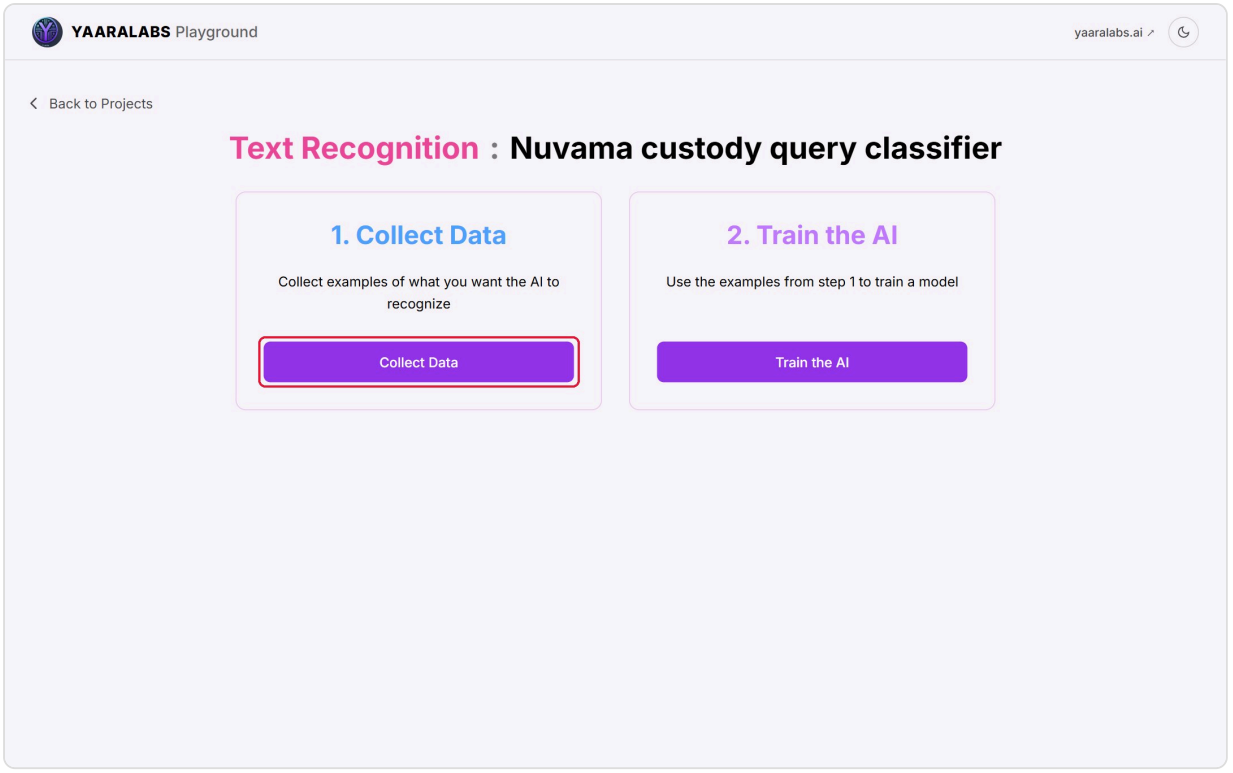
1 Create the project

Type a project name, choose “Text Recognition”, then click Create Project.



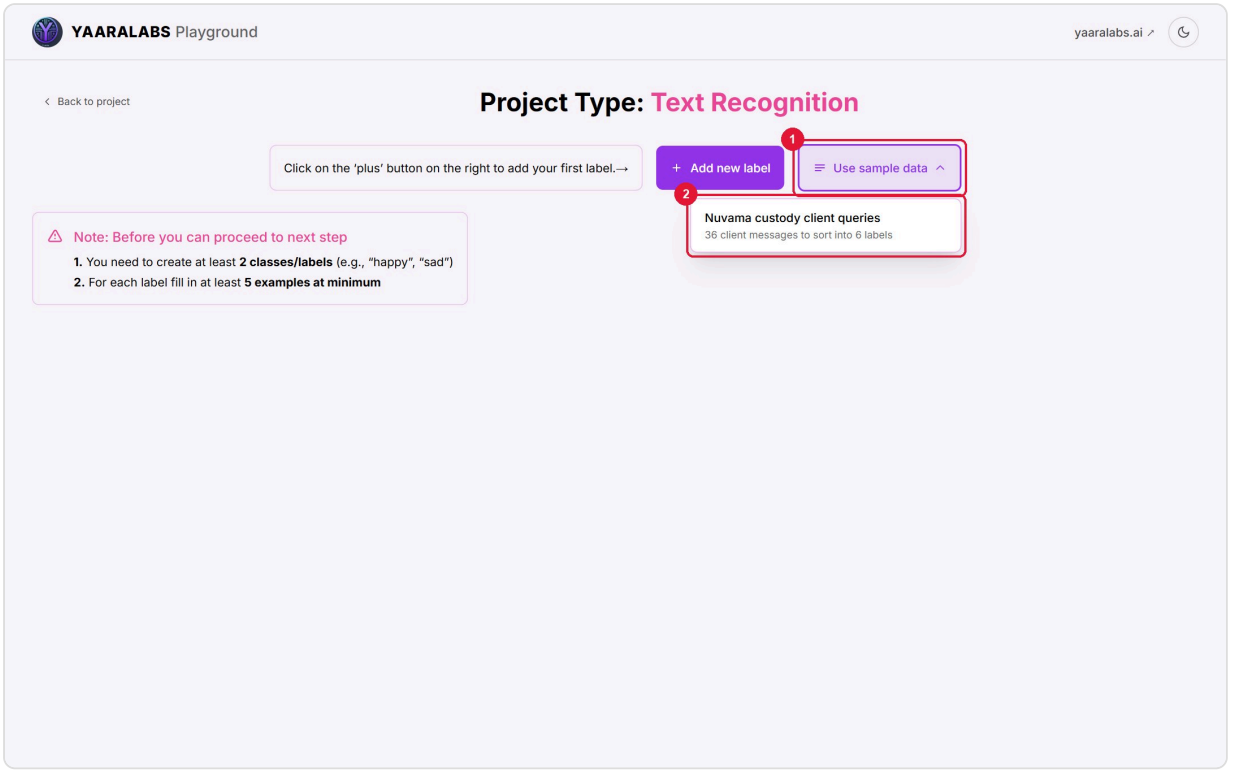
2 Open Collect Data

The project has three stages: collect data, train, use. Start with Collect Data.



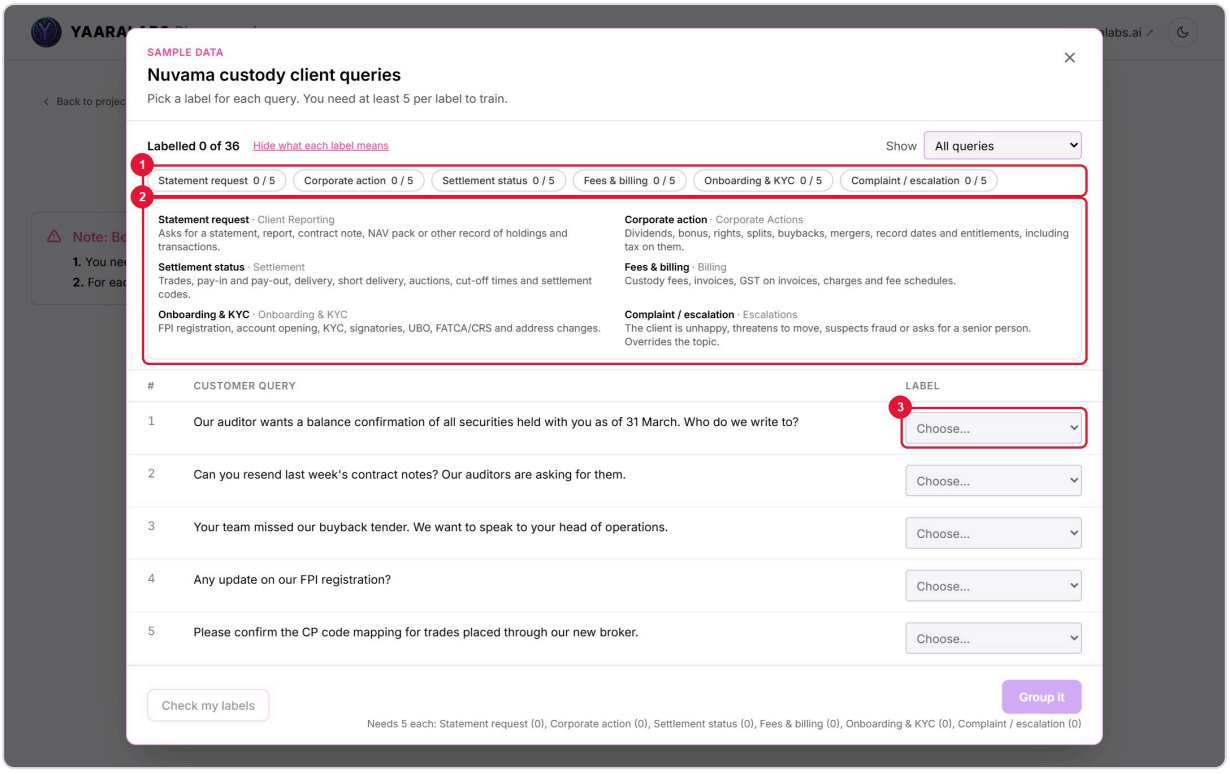
3 Load the sample data

Instead of typing labels by hand, open “Use sample data” and pick “Nuvama custody client queries”.



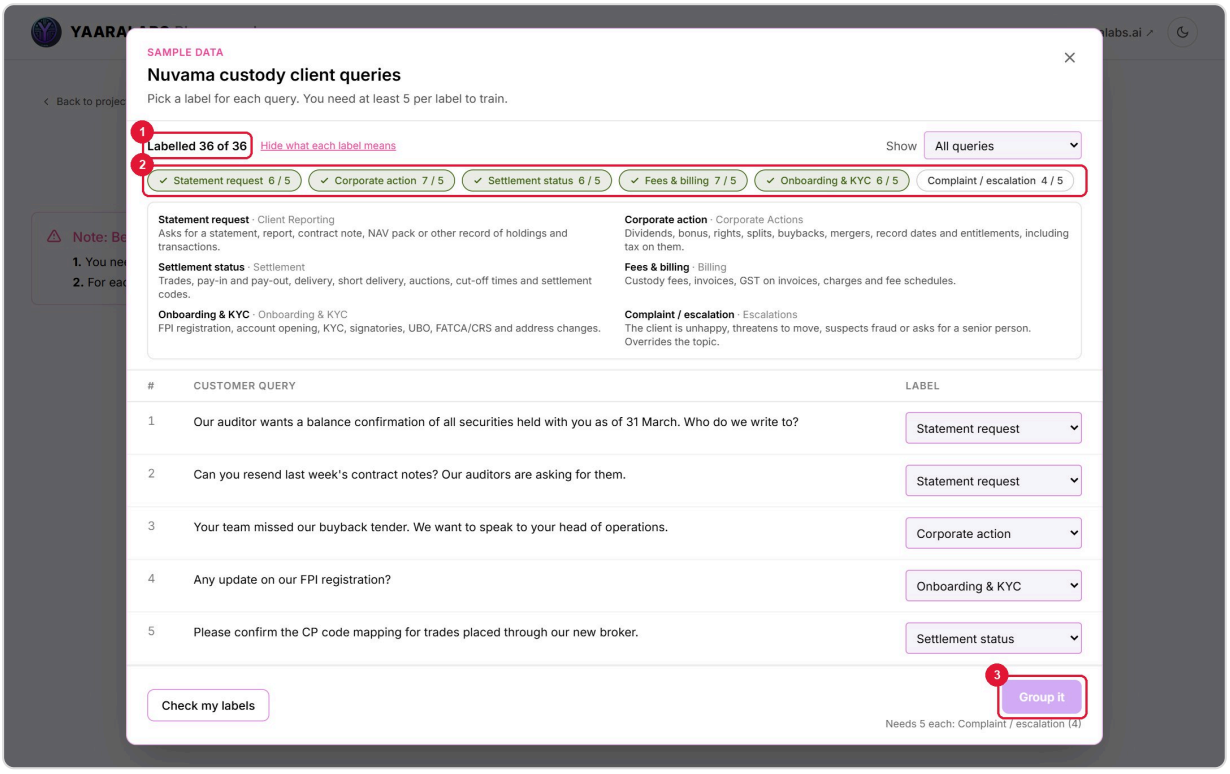
4 The labelling table

36 real-style client queries, none labelled. The six labels and what each covers are shown; the counters track 5 needed per label.



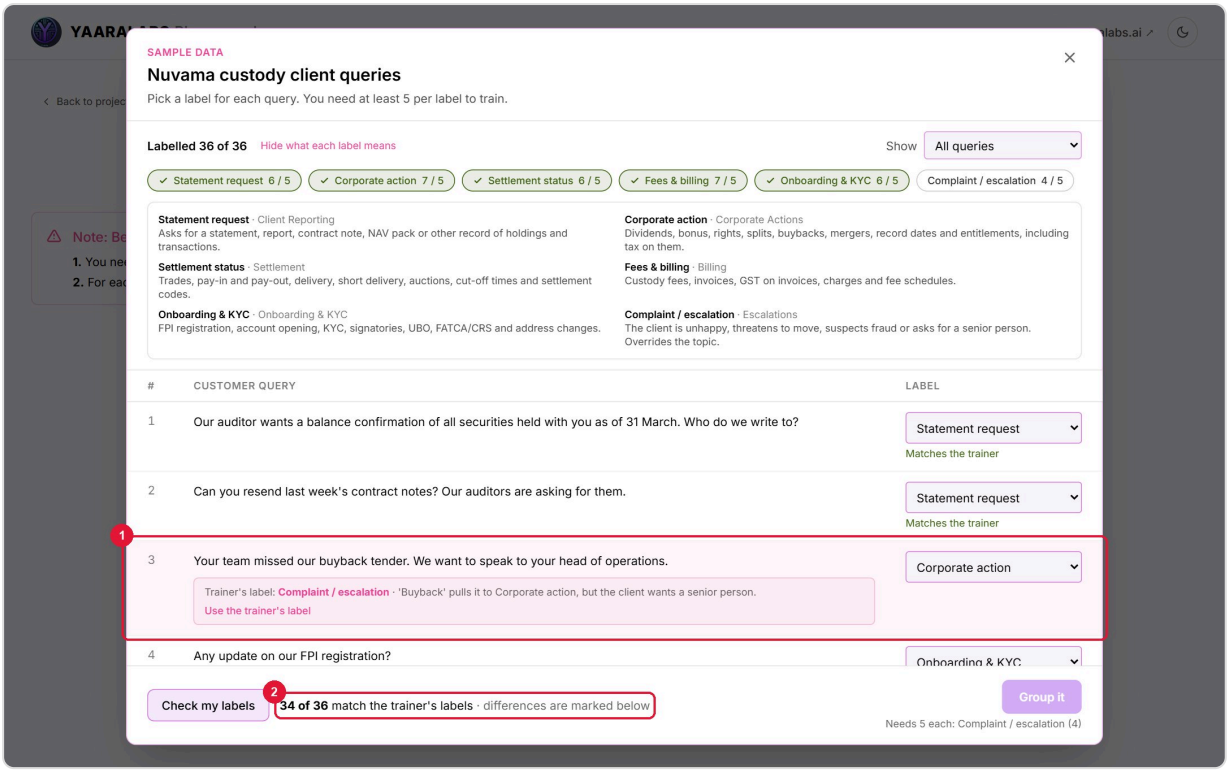
5 Label every query

Each query gets a label from its dropdown. All 36 labelled, every counter is green, so Group it is enabled.



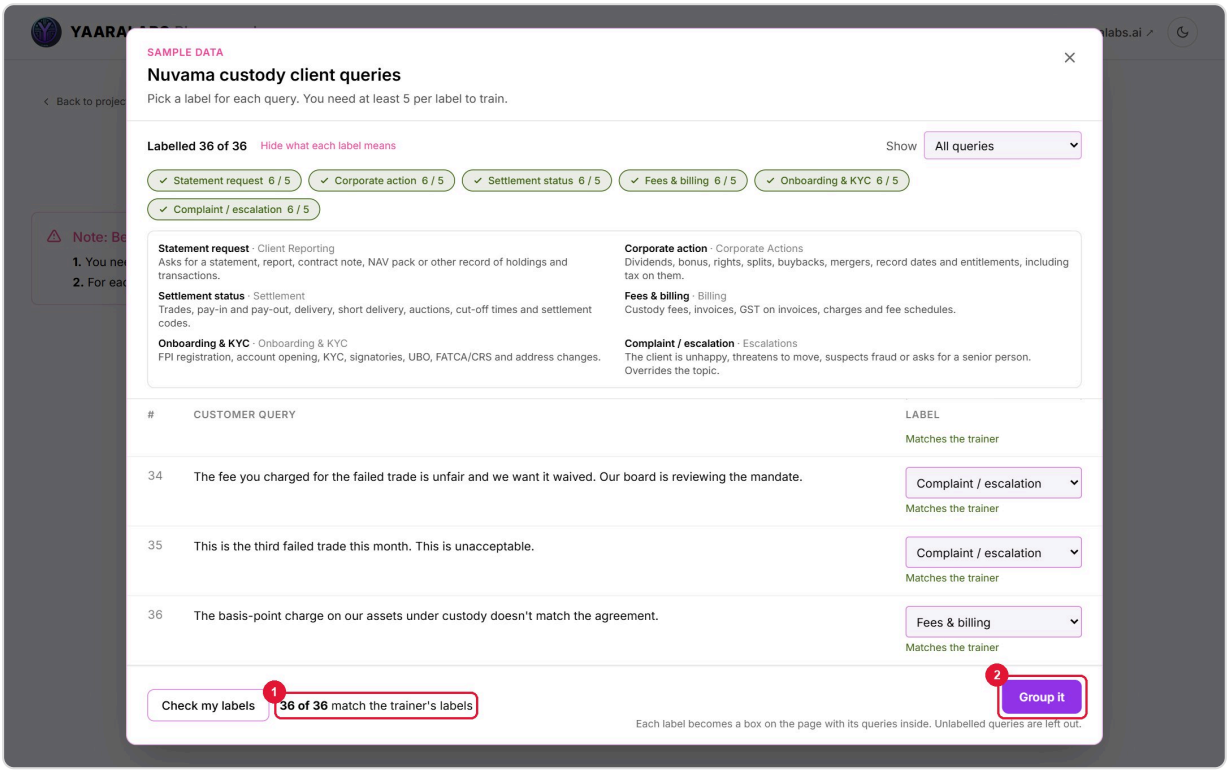
6 Check my labels

"Check my labels" compares with the trainer: 34 of 36 match. Row 3 mentions a buyback, but the client wants a senior person, so it is a complaint.



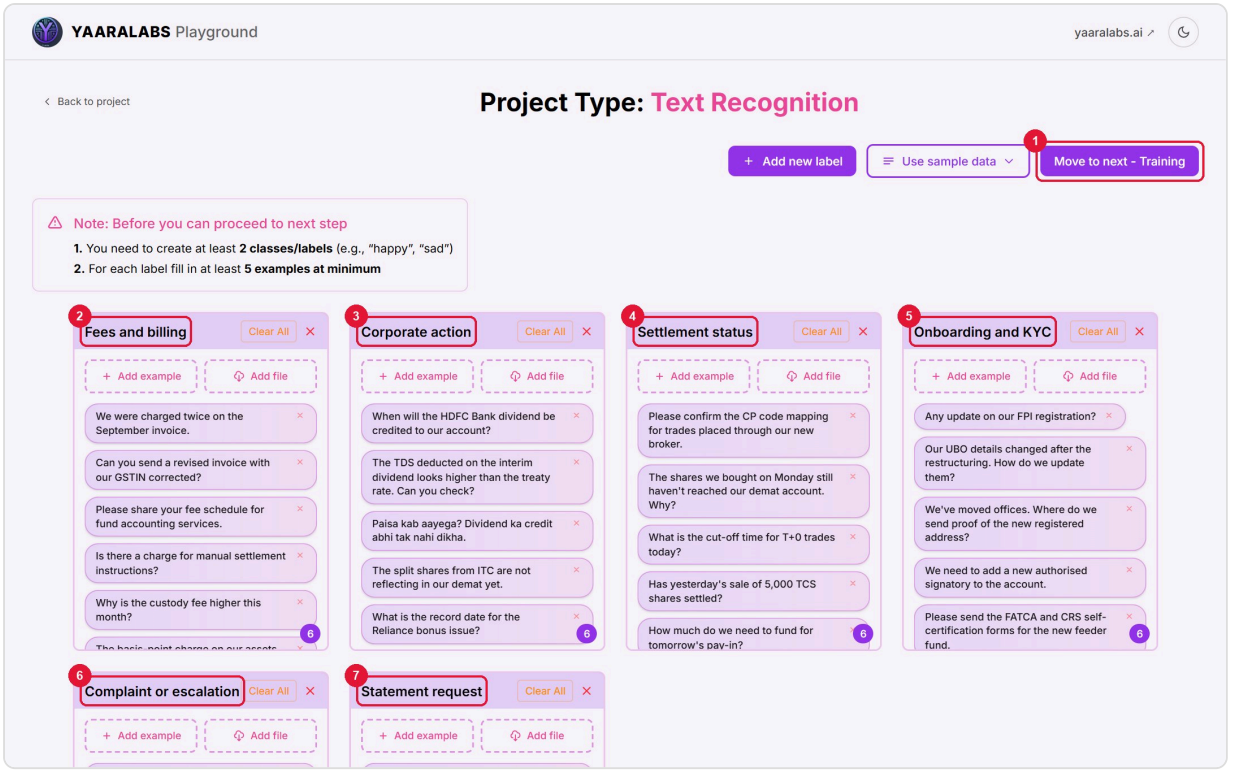
7 Fix the differences, then Group it

Accept the trainer's label on the two rows: 36 of 36 now match. Click Group it.



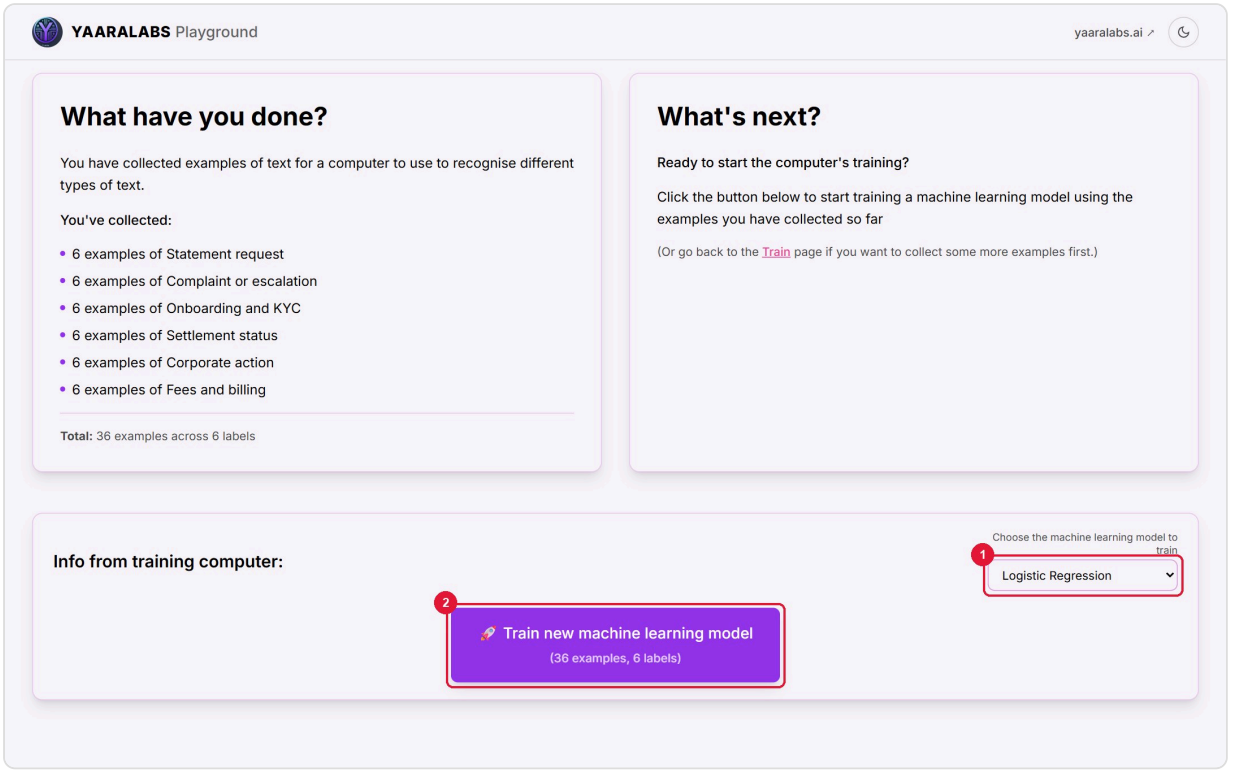
8 Each label becomes a box

Group it creates one box per label with its queries inside, just like labels added by hand. Move to Training.



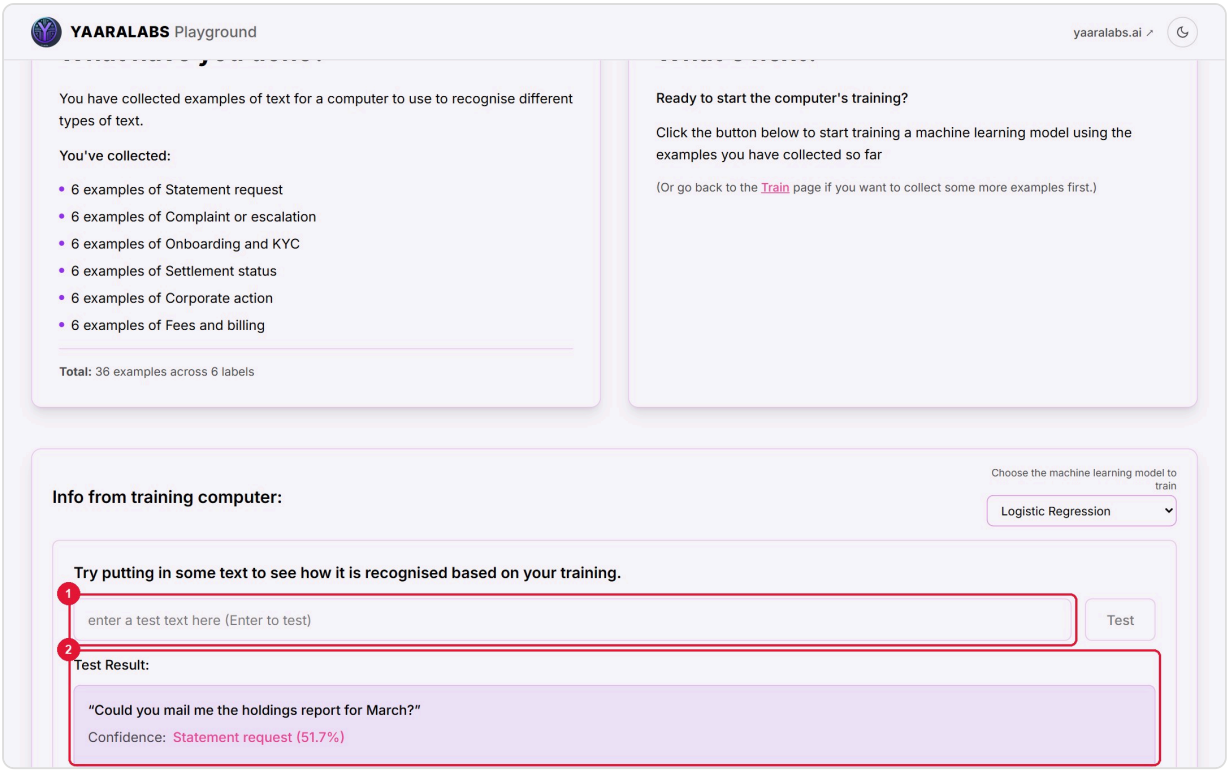
9 Choose the model and train

The model list offers Logistic Regression and DistilBERT. Choose Logistic Regression and click Train (36 examples, 6 labels).



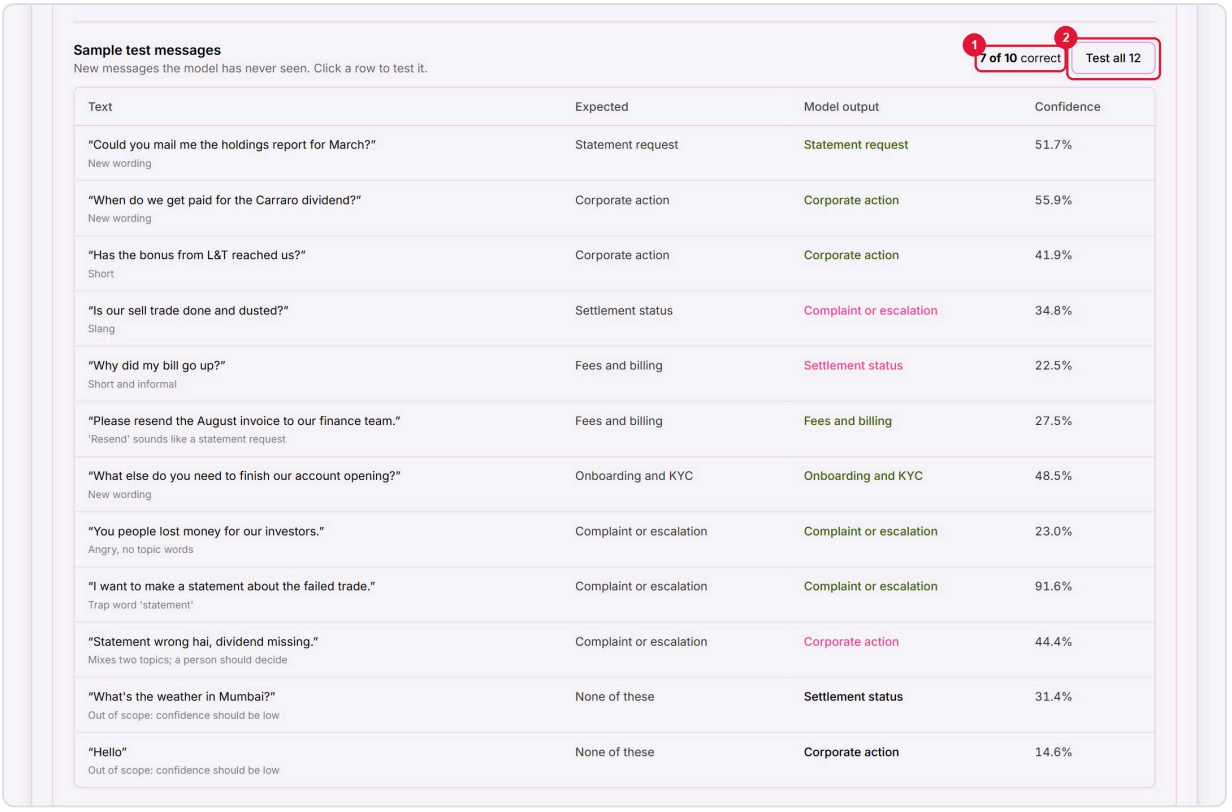
10 Try the model

Type any client message and click Test: the model returns a label and how confident it is.



11 Results on the 12 test messages

Open Test data and click Test all 12: each unseen message shows the expected label, the model's label and its confidence.



Text	Expected	Model output	Confidence
"Could you mail me the holdings report for March?" New wording	Statement request	Statement request	51.7%
"When do we get paid for the Carraro dividend?" New wording	Corporate action	Corporate action	55.9%
"Has the bonus from L&T reached us?" Short	Corporate action	Corporate action	41.9%
"Is our sell trade done and dusted?" Slang	Settlement status	Complaint or escalation	34.8%
"Why did my bill go up?" Short and informal	Fees and billing	Settlement status	22.5%
"Please resend the August invoice to our finance team." 'Resend' sounds like a statement request	Fees and billing	Fees and billing	27.5%
"What else do you need to finish our account opening?" New wording	Onboarding and KYC	Onboarding and KYC	48.5%
"You people lost money for our investors." Angry, no topic words	Complaint or escalation	Complaint or escalation	23.0%
"I want to make a statement about the failed trade." Trap word 'statement'	Complaint or escalation	Complaint or escalation	91.6%
"Statement wrong hai, dividend missing." Mixes two topics; a person should decide	Complaint or escalation	Corporate action	44.4%
"What's the weather in Mumbai?" Out of scope: confidence should be low	None of these	Settlement status	31.4%
"Hello" Out of scope: confidence should be low	None of these	Corporate action	14.6%

Results on the 12 unseen test messages (Logistic Regression)

Logistic Regression

7 of 10 in-scope messages correct

Out-of-scope messages

31%, 15% confidence (low, as it should be)

Test message	Expected	Model said	Confidence	Result
Could you mail me the holdings report for March? New wording	**Statement request**	Statement request	51.7%	Correct
When do we get paid for the Carraro dividend? New wording	**Corporate action**	Corporate action	55.9%	Correct
Has the bonus from L&T reached us? Short	**Corporate action**	Corporate action	41.9%	Correct
Is our sell trade done and dusted? Slang	**Settlement status**	Complaint or escalation	34.8%	Wrong
Why did my bill go up? Short and informal	**Fees and billing**	Settlement status	22.5%	Wrong
Please resend the August invoice to our finance team. 'Resend' sounds like a statement request	**Fees and billing**	Fees and billing	27.5%	Correct
What else do you need to finish our account opening? New wording	**Onboarding and KYC**	Onboarding and KYC	48.5%	Correct
You people lost money for our investors. Angry, no topic words	**Complaint or escalation**	Complaint or escalation	23.0%	Correct
I want to make a statement about the failed trade. Trap word 'statement'	**Complaint or escalation**	Complaint or escalation	91.6%	Correct
Statement wrong hai, dividend missing. Mixes two topics; a person should decide	**Complaint or escalation**	Corporate action	44.4%	Wrong
What's the weather in Mumbai? Out of scope: confidence should be low	**None of these**	Settlement status	31.4%	Low confidence: good
Hello Out of scope: confidence should be low	**None of these**	Corporate action	14.6%	Low confidence: good

Conclusion

WHAT THE EXPERIMENT SHOWS

- The model learned intents from 36 labelled examples, with no rules written.
- It handles new wording the training data never contained, which rules could not.
- A confidence score tells you when to let a person decide.

WHAT TO WATCH

- Six examples per intent is very little: short, slangy or mixed messages are still missed.
- Label quality matters: the “Check my labels” step catches intent-vs-keyword mistakes.
- The model always picks a label, so out-of-scope messages need a low-confidence rule.

Takeaway: from just 36 labelled messages, Logistic Regression sorted 7 of 10 unseen client messages correctly, including the trap “I want to make a statement about the failed trade” at 92% confidence, which keyword rules got wrong in Experiment 1. More labelled examples make it better; Experiment 3 shows exactly where a model goes wrong, using a confusion matrix.